Vincent Lam

Software Engineering | Security & AI Agents | Open to remote

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Software engineer (B.S. Computer Science, 2025) who ships full-stack and AI-agent projects end to end: design or threat model, code, automated tests, CI gates and a live demo. Security-minded by default (least privilege, policy-as-code, LLM features that resist prompt injection), with 2+ years of accuracy-critical work for the State of California.

SKILLS

Identity & access: Least privilege, ABAC/RBAC, Cedar policy-as-code, SCIM 2.0, access reviews, OIDC/JWT, MFA, AWS IAM & STS
AI & agents: Claude API, tool use, structured outputs, prompt injection defense, agent permissioning, Pydantic / Zod
Security: STRIDE threat modeling, NIST 800-53 / SOC 2 mapping, audit logging, detection rules, SIEM (OCSF), CodeQL, Bandit
Languages & tools: Python, TypeScript, JavaScript, SQL, Java, R, FastAPI, Next.js, React, Docker, Git, GitHub Actions, pytest, Vitest

PROJECTS (2026)

Aegis – Just-in-time access platform, zero standing privilege | Python, FastAPI, Cedar, OIDC, SCIM, AWS STS — Live demo | Code

- **Removed standing access.** Cedar policies decide each request on user and resource attributes; restricted access needs MFA step-up and a second approver; approved grants become scoped AWS STS credentials that expire on their own and are revoked on SCIM 2.0 joiner/mover/leaver events.
- **Made an LLM safe to put in front of access decisions.** Claude only extracts fields from untrusted text, injection detection sends manipulated requests to a human, and tests with a hijacked parser confirm policy still decides.
- **Made every decision auditable and shipped it like production code.** HMAC-chained audit log with signed checkpoints, 6 detection rules with OCSF SIEM export, a STRIDE threat model and a NIST 800-53 / SOC 2 map; 186 tests at 93% coverage gated in CI by CodeQL, Bandit, mypy and pip-audit, plus a live demo that runs the real Cedar engine in the browser.

Nagare – Japanese reader that teaches as you read | TypeScript, Next.js, React, IndexedDB — Live demo | Code

- **Built a reader with no backend.** Imports .txt, .epub and .pdf, splits Japanese into words in the browser (kuromoji.js) and looks up about 22,000 JMdict words tagged by JLPT level; progress and spaced-repetition flashcards live in IndexedDB. A script levels about 30 classic novels by their vocabulary, and a proofreading pass fixed 222 wrong word meanings.

Kumo – Claude Code agent on a home NAS, driven from Telegram | Docker, Bash, TypeScript, Claude API — Write-up

- **Treated the agent as a privileged identity.** Threat-modeled it first, then bounded it: one allow-listed user, no root, no open ports, destructive commands behind one-tap approvals, and scripts it writes run in a no-network Docker container.
- **Built a Japanese-learning companion on it.** Replies stay within the learner's known vocabulary, weak words return on a spaced-repetition schedule inside normal chat, and prompt caching halved reply cost; 275 tests run against a fake model.

Privacy Blocker – Local-first data-broker opt-out tool | JavaScript, Web Crypto, GitHub Actions — Live demo | Code

- **Built so it cannot leak what it protects.** Data is encrypted in the browser (PBKDF2-SHA256, then AES-256-GCM) and a CSP of connect-src 'none' blocks outbound requests; a scheduled GitHub Actions job sends CCPA deletion requests to ~25 brokers per run.

Installous – AI stock research agent | TypeScript, Next.js, Claude API — Live demo | Code

- **Grounded the AI advisor in live data and tested the model in the open.** A Claude advisor reads quotes and news through Zod-validated tools; a rules-only $100k paper fund trades a five-factor score on a schedule, logs each trade's reason and is benchmarked against SPY.

W.E.M.U 2.0 – Senior project, team of 8, for the CA Dept. of Consumer Affairs | Node.js, Selenium

- **Moved licensing forms to a new system at scale.** Co-built the form migration for 3.4M licenses across 36 entities; wrote back-end logic for the site's button actions and owned the production stage, contact forms page and Selenium role-based access tests.

EXPERIENCE

Office of Legislative Counsel – *Office Technician (Typing)* — *Sep 2026 – Present*

- Produce and publish legislation in document-workflow systems on a 5-person team, delivering finals through secure channels.

California Department of Health Care Services

Student Assistant – Public Hospital Redirection Payments & Veterans Homes Programs — *Jan 2025 – Jun 2025*

- **Caught and resolved discrepancies before payment release.** Reconciled thousands of claims and financial reports per cycle from hospitals, doctors and other providers statewide for state and federal reimbursement programs, with fiscal and policy stakeholders.

Student Assistant – SharePoint Program & Records Support — *Nov 2022 – Apr 2024*

- **Replaced manual document sends.** Helped build a Power Automate flow for the monthly, quarterly and annual sends; handed it off.
- **Gave staff one place to find liaisons and kept SharePoint healthy.** Built the landing page on the Outlook Global Address List (access limited to designated staff); resolved access issues and broken links, and QA'd accuracy, metadata and accessibility.

California Housing Finance Agency – *Student Assistant, Loan Operations & Data Quality* — *May 2022 – Oct 2022*

- **Sped up loan-data review** by adding features to 3+ existing Python and R data-validation scripts.

EDUCATION

California State University, Sacramento – B.S. Computer Science — Aug 2025
Coursework: Computer System Attacks & Countermeasures, Computer Networks, Operating Systems, Databases, Data Mining